

IGIS
INSTITUTE OF GEOGRAPHICAL INFORMATION SYSTEMS

INTERNATIONAL TRAINING WORKSHOP

Remote Sensing *for Disaster Management*

From satellite constellations to UAV systems: an integrated observation chain for emergency response and loss assessment

18–20 November 2026
IGIS, SCEE, NUST · Islamabad

Programme booklet

Two scales. One observation chain.

During a monsoon flood, responders need two answers at once: *how far has the water spread, and who is affected?* Then: *which embankments breached, which bridges failed, how many homes are lost?*

> No single sensing platform can address both. Used in a deliberate sequence, satellites and UAVs create an observation chain far more effective than either alone.

Over two and a half days you will work through that chain, using openly licensed satellite data, browser-based cloud computing and standard commercial UAVs, all reproducible with equipment ordinarily available to a Pakistani institution.

BY THE CLOSE OF DAY 3 YOU WILL HAVE PRODUCED

1. **A flood extent map** from Sentinel-1 radar
2. **An impact table** of population, cropland and roads
3. **A drone map and 3D model** from a survey you helped fly
4. **A damage map** read from drone imagery
5. **A rapid-response brief** from the capstone exercise

PROGRAMME AT A GLANCE

From the wide view to the close-up.

01

WEDNESDAY 18 NOVEMBER · 09:15–16:00

Satellite remote sensing for floods

Modules 1–3 · Lab 1 flood mapping from radar · Lab 2 who and what was affected

02

THURSDAY 19 NOVEMBER · 09:15–16:00

UAV remote sensing: from flight to damage map

Module 4 · flight planning · field flights · Lab 3 photos to maps · Lab 4 reading damage from the air

03

FRIDAY 20 NOVEMBER · 09:15–12:15, CLOSING 14:30

Integration, case studies & rapid response

Module 5 · three Pakistani case studies · capstone simulation · closing ceremony after Friday prayers

2½	13	30	4
days	contact hours	participants	expert lecturers

DAY 1 · WEDNESDAY 18 NOVEMBER

Satellite remote sensing for floods

MORNING

09:00	**Registration and tea** Badge collection, laptop and Earth Engine login check.
09:15	**Opening ceremony** Welcome address, inauguration and course objectives.
09:35	**Module 1 · Seeing disasters from space** What satellites can and cannot see; optical versus radar imagery; resolution and revisit time; satellites across the disaster management cycle.
10:20	**Module 2 · Free satellite data for floods** Sentinel-1, Sentinel-2, Landsat and rainfall data, free through Copernicus and Earth Engine; requesting emergency maps through the International Charter.
11:00	Tea break
11:15	**Module 3 · Pakistan's floods seen from space** The 2010 and 2022 floods: what satellite maps showed responders and where they fell short.
12:00	Lunch and prayers

DAY 1 · WEDNESDAY 18 NOVEMBER

Mapping the flood and its impact

AFTERNOON

13:30	**Lab 1 · Map a flood from radar**	Short demonstration, then run a ready-made script on before-and-after Sentinel-1 images to map the 2022 Sindh flood extent. No coding needed.
14:30	Tea break	
14:45	**Lab 2 · Who and what was affected?**	Overlay the flood map on population, cropland and roads with a provided script, and read the impact table for one district.
15:30	**Wrap-up and Q&A**	Recap of the day and a preview of the drone day.
16:00	Close of Day 1	

DAY 2 · THURSDAY 19 NOVEMBER

UAV remote sensing: from flight to damage map

09:00	Arrival and tea
09:15	**Module 4 · UAV systems and the rules** Drone types and sensors; regulations and flight safety in Pakistan.
10:00	**Flight planning exercise** In groups, plan the mission you will fly: area, height, overlap and resolution.
10:30	Transfer to the NUST field site Near the Saddle Club.
10:45	**Field practical · UAV flights** Safety briefing and demo flight, then groups fly their planned mission under supervision.
12:15	Data offload, return to IGIS, lunch and prayers
13:45	**Lab 3 · From photos to maps** Turn drone photos into a stitched map and a 3D model, using a prepared dataset.
14:45	Tea break
15:00	**Lab 4 · Reading damage from the air** Spot damage by eye in the drone map and 3D model, then compare with the satellite view.
15:40	**Wrap-up and Q&A**

DAY 3 · FRIDAY 20 NOVEMBER

Integration, case studies & rapid response

09:00	Arrival and tea
09:15	**Module 5 · Putting satellites and drones together** Satellites to see the whole event and decide where to look; drones for detail in priority areas; field checks to confirm.
09:45	**Applied case studies** Attabad Lake flood risk · Shisper Lake outburst flood scenarios · Jhang riverine flooding and the monsoon.
10:20	Short break Scenario packs handed out.
10:30	**Capstone · Rapid-response simulation** Groups take a disaster scenario and prepare a short brief: a satellite map of the affected area, who and what was affected, and where they would send a drone.
11:30	**Group presentations** Each group presents, with feedback from the instructors.
12:15	Friday prayers and lunch
14:30	**Closing ceremony** Course summary, remarks by dignitaries, certificates and group photograph.

LEARNING OUTCOMES

What you'll be able to do.

01 Explain what satellites can and cannot see, and optical versus radar imagery.

02 Find free satellite data for floods and request international emergency maps.

03 Map flood extent from Sentinel-1 radar with a ready-made workflow.

04 Estimate the population, cropland and roads affected by a flood.

05 Describe drone types and sensors, and Pakistan's rules for flying safely.

06 Plan a simple drone survey and take part in a supervised flight.

07 Turn drone photos into a stitched map and a 3D model.

08 Recognize damage in drone imagery and what it adds to the satellite view.

09 Combine satellite and drone information into a rapid-response brief.

EXPERT LECTURERS

Your lecturers.



Dr. NIE Yong

Professor · Institute of Mountain Hazards and Environment (IMHE), CAS

Director, Mountain Science Data Center. Glacier and glacial lake hazards and remote sensing; 50+ papers including ESI highly cited work; risk assessment for the China–Pakistan Economic Corridor.



Engr. WANG Wen

Assistant Engineer · Mountain Science Data Center, IMHE, CAS

Remote sensing for mountain hazards and the cryosphere; extensive UAV fieldwork and processing, from flight planning to orthomosaics, DEMs and 3D models.



Dr. Javed Iqbal

Professor · SCEE – IGIS, NUST

Three decades in GIS, remote sensing and hazard mapping of landslides, glacier avalanches and flash floods; 200+ scholarly contributions.



Dr. Shahid Nawaz Khan

Assistant Professor · SCEE – IGIS, NUST

Explainable machine learning for Earth observation, SAR–optical fusion, Earth Engine and UAV remote sensing; FAA Part 107 remote pilot.

BEFORE YOU ARRIVE

Come ready to work.

The schedule is compressed, so there is no slack to recover from a failed login on the day.

- [ ] **Google Earth Engine access** registered and verified with a linked Cloud project. Allow 7–10 days.
- [ ] **Laptop with Google Chrome.**
- [ ] **QGIS 3.34 LTR** installed, with 5 GB free disk space.
- [ ] **UAV ground-station app** on a charged phone or tablet, with a cable.
- [ ] **Field clothing for Day 2:** closed-toe shoes, sun protection and a hat.
- [ ] **Pre-reading pack**, circulated two weeks ahead.
- [ ] **Signed field-safety acknowledgement**, handed in at registration.

VENUE & CONTACT

Where to find us.

LECTURES & LABS · ALL THREE DAYS

Institute of Geographical Information Systems

SCEE, NUST Sector H-12, Islamabad 44000

UAV FLIGHTS · DAY 2 MORNING

NUST Field Site

Near the Saddle Club, NUST H-12 · transfer from IGIS at 10:30

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NOTES

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IGIS
INSTITUTE OF GEOGRAPHICAL INFORMATION SYSTEMS



China-Pakistan Joint Research Center on Earth Sciences (CPJRC)
Institute of Geographical Information Systems, School of Civil and Environmental Engineering, NUST
Geological Survey of Pakistan

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